ADAMAS UNIVERSITY

FAST FOOD

Comprehensive Project Documentation

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Project Overview

FAST FOOD is a modern web-based food delivery and restaurant management platform designed to bridge the gap between customers and local eateries. The application provides a seamless ordering experience for users while offering a robust management suite for administrators.

In today's fast-paced world, efficiency and speed are paramount. Our solution focuses on performance, role-based security, and real-time data visualization to ensure that both the consumer and the business owner have all the tools they need at their fingertips.

Technology Stack

Frontend

Built with efficiency and accessibility in mind.

- HTML5
- CSS3 (Vanilla)
- JavaScript (ES6+)
- Chart.js
- Google Fonts

Backend

A lightweight and scalable server architecture.

- Python
- Flask
- Flask-CORS
- RESTful APIs

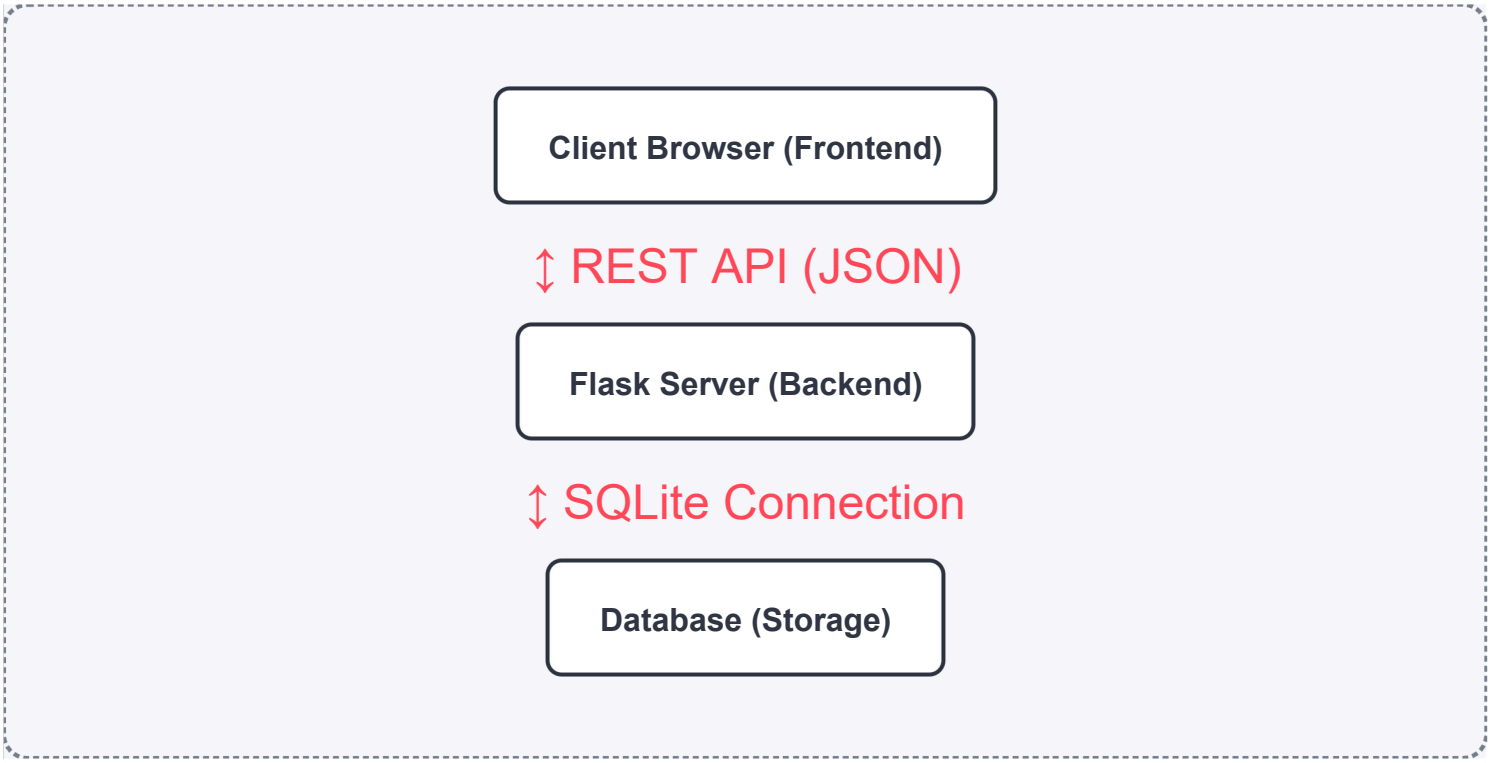
Database

Structured and reliable data persistence.

- SQLite
- SQL Queries
- Relational Mapping

System Architecture

The system follows a classic Client-Server architecture with a clean separation between the presentation layer and the data processing layer.



Data Flow

User Action: User interacts with the UI (e.g., adding to cart).
API Request: Frontend sends an asynchronous request (Fetch API) to the Flask backend.
Processing: Backend validates tokens/roles and executes SQL queries.
Response: Structured JSON is returned to the frontend.
Rendering: UI updates dynamically without full page reloads.

Key Features

1. Role-Based Access Control (RBAC)

The application features a secure login system that differentiates between standard users and system administrators. Users are redirected to the ordering portal, while admins have access to the Dashboard and Analytics.

2. User Ordering Portal

Restaurant Discovery

Dynamic listing of all registered restaurants with ratings and locations.

Smart Shopping Cart

Real-time total calculations and multi-item management before order placement.

Order History

Personalized view of all past orders with status tracking (Pending/Delivered).

3. Admin Management Suite

Admins can perform full CRUD (Create, Read, Update, Delete) operations on core entities:

Menu Management: Update prices, descriptions, and dish images.
Restaurant Management: Register new locations or modify existing ones.

User Management: Securely add and monitor platform users.

4. Advanced Analytics

Integrated Data Visualization using Chart.js allows admins to monitor:

Popular Items: Visualization of the most ordered dishes across the platform.
Revenue Streams: Comparative analysis of restaurant performance.
Sales Timeline: Time-series analysis of orders for specific restaurants.

Database Schema

The relational database is designed for integrity and performance. Key tables include:

Table	Columns	Description
`users`	id, name, email, phone	Stores registered platform users.
`restaurants`	id, name, location, rating	Entities hosting food menus.
`menu`	id, restaurant_id, item_name, price, image_url	Food items linked to restaurants.
`orders`	id, user_id, status, total_amount, created_at	Transactional order records.
`order_items`	id, order_id, menu_id, quantity	Junction table for order details.

Installation & Setup

Follow these steps to deploy the application in a local development environment:

Prerequisites

Python 3.8 or higher
Modern Web Browser (Chrome/Edge recommended)
Node/NPM (Optional, backend is pure Python)

Step-by-Step Guide

Clone the Project: Navigate to the project root directory.
Setup Virtual Environment:

```
python -m venv venv
source venv/bin/activate  #Windows: venv\Scripts\activate
```

Install Dependencies:

```
pip install -r backend/requirements.txt
```

Initialize Database:

```
python backend/db.py
```

Launch Application: Run the `start.bat` file or manually launch the backend and open `frontend/index.html`.

Conclusion

The FAST FOOD project successfully demonstrates a full-stack solution for the food delivery industry. By leveraging modern technologies like Flask and Chart.js, we have created a platform that is not only functional but also provides valuable business insights.

Thank you for reviewing our project.